Oil & Gas Data Engineering Project

Microsoft Fabric - Project Documentation

Project Overview

An end-to-end data engineering project built in Microsoft Fabric using a real Brazilian oil and gas production dataset. The goal is to ingest raw production data, clean and transform it, organize it into analytical layers, build a dimensional data model, and produce a concise report for production, field, well, and injection analysis.

Data Source

Source	Kaggle - Brazil oil and gas production
File	brazil_oil_production.sqlite (SQLite database)
Coverage	Monthly production data, 1954-2020
Dataset URL	https://www.kaggle.com/datasets/luciodias/brazil-oil-production

Architecture & Data Flow

ELT approach: the raw SQLite data is loaded into the Fabric Lakehouse first, then cleaned and transformed inside Microsoft Fabric. The project follows a Medallion Architecture: **Bronze** stores the raw source data, **Silver** contains cleaned and standardized data, and **Gold** contains analysis-ready fact and dimension tables.

Microsoft Fabric Components

OneLake / Lakehouse: data storage **Data Pipeline:** workflow orchestration **Notebook + PySpark:** cleaning and transformations **Semantic Model:** relationships and analytical model **Report:** final visualization and analysis.

Data Cleaning & Transformation

The processing includes checking missing values and duplicates, correcting invalid data types, coercing malformed numeric values, and handling missing values according to their meaning. For example, missing descriptive installation values are represented as "Not Reported", while sparse injection measurements are not automatically converted to zero because a missing value does not necessarily mean zero activity.

Gold Layer & Data Model

The final analytical model uses a Star Schema. The main table is **gold_fact_production**, connected through one-to-many relationships to **gold_dim_date**, **gold_dim_field**, and **gold_dim_well**. This structure supports filtering and analysis by time, field, well, state, basin, and production metrics.

Automation & Validation

The Data Pipeline executes the project notebook and was validated with a successful run. A daily schedule was also configured so the pipeline can run automatically, demonstrating an automated refresh mechanism for the data workflow.

Final Report

The final report contains three focused pages: **Production Overview**, **Field & Well Analysis**, and **Injection & Operations Analysis**. It summarizes key production KPIs, trends over time, top producing fields, geographic production, and injection activity.

Project workflow: Source Data -> Bronze -> Silver -> Gold -> Star Schema / Semantic Model -> Report